Advanced Mukai classification from independent quantum correspondences

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Abstract

We deduce the advanced generalized Mukai classification from two stipulated inputs in a pinned source: the generalized Mukai inequality with its equality case, and an internal proposition giving a nonzero product of quantum correspondences in independent numerical curve degrees. The second input is essential and is stronger than the inequality alone. Retaining the dimension of the supported composed cycle gives a closed rational-chain relation of dimension at least $2\dim X - 1$. A saturated quotient and a divisor argument then give chain connectedness. The resulting bounded collection supplies competitors at every minimal horizontal step of Fujita's construction, establishing its unsplitness hypothesis without restrictions on dimension or Picard number. Fujita's existing induction and bundle classifications finish the proof. The source inputs are assumed, not proved or audited here.

1 Statement, dependencies, and contribution

All varieties are over $\mathbb{C}$. A Fano manifold is a smooth connected projective variety X with $-K_X$ ample. Write $n = \dim X > 0$, $r = \rho(X)$ for its *actual* Picard number, and

$$i = \iota(X) = \min\{-K_X \cdot C : C \subset X \text{ an irreducible rational curve}\}.$$

Projectivizations parametrize one-dimensional quotients. A product with zero factors is a point. Numerical curve spaces have real coefficients.

The argument uses the following two assertions of the source [1], at the fixed revision recorded in Section 8. They are stipulated throughout, including when applied to a Fano base of smaller Picard number.

S1. Every positive-dimensional Fano manifold satisfies $r(i-1) \le n$, with equality precisely for $(\mathbb{P}^{i-1})^r$.

S2. If $i \ge 2$, there are r linearly independent numerical stable-map degrees $\gamma_1, \dots, \gamma_r$, with $d_j = -K_X \cdot \gamma_j \ge i$, such that

$$S_{\gamma_1} \cdots S_{\gamma_r} \neq 0, \qquad \sum_{j=1}^{r} (d_j - 1) \le n. \tag{1}$$

Here S_γ acts on even cohomology and lowers complex codimension by $d_\gamma - 1$. It is induced by the algebraic two-evaluation pushforward of the genus-zero virtual class, of cycle dimension $n + d_\gamma - 1$ on $X \times X$. For a numerical degree the source sums the finitely many actual curve classes.

S1 is the source's main inequality and equality theorem. S2 is an internal proposition about independent correspondences. We do not assert that it follows from S1. Its spectral and descendant inputs are also stipulated. Thus the following theorem is a conditional deduction with these exact dependencies.

Theorem 1.1. *Assume S1 and S2. If a positive-dimensional Fano manifold X satisfies $r(i-1) \geq n-1$, it is isomorphic to one of the following varieties:*

(1) $(\mathbb{P}^{i-1})^r$;

(2) $Q^i \times (\mathbb{P}^{i-1})^{r-1}$, *where* $i \geq 3$;

(3) $\mathbb{P}_{\mathbb{P}^i}(\mathcal{O}^{\oplus(i-1)} \oplus \mathcal{O}(1)) \times (\mathbb{P}^{i-1})^{r-2}$, *where* $r \geq 2$ *and* $i \geq 2$;

(4) $\mathbb{P}_{\mathbb{P}^i}(T_{\mathbb{P}^i}) \times (\mathbb{P}^{i-1})^{r-2}$, *where* $r \geq 2$ *and* $i \geq 2$;

(5) $\mathbb{P}^i \times (\mathbb{P}^{i-1})^{r-1}$, *where* $r \geq 2$ *and* $i \geq 2$.

Here Q^i is a smooth quadric of dimension i. Every variety listed with these restrictions is Fano with actual Picard number r and pseudoindex i. Type (1) *has $n = r(i-1)$; types* (2)–(5) *have $n = r(i-1)+1$.*

The hypothesis forces $i \geq 2$: otherwise $n \leq 1$, whereas a Fano curve is $\mathbb{P}^1$ and has pseudoindex two. There is no extra hypothesis about a contraction, symmetry, or covering family. At $r = 1$ the list consists of $\mathbb{P}^n$ and, for $n \geq 3$, Q^n. The quadric $Q^2 \simeq \mathbb{P}^1 \times \mathbb{P}^1$ has actual Picard number two and lies in type (1). In dimensions one and two the possibilities are respectively $\mathbb{P}^1$, and $\mathbb{P}^2$ or $\mathbb{P}^1 \times \mathbb{P}^1$. When $i = 2$, the defect-one list has just types (3)–(5). The restriction $r \geq 2$ in type (5) matters: $\mathbb{P}^i$ alone has pseudoindex $i+1$.

Fujita formulated this five-type classification and proved its pseudoindex version for Picard parameter at most three or dimension at most five [2, Conjecture 1.4, Proposition 1.8, Remark 1.9], using the rank-one input. We use Dedieu–Höring's established quadric characterization [3, Theorem 1.3] for that input. The latter paper addresses the gap in the older rank-one argument. Fujita's minimal-family construction, quotient theorem, extremal-ray criterion, induction, and bundle classifications remain external ingredients here.

The addition is the integrated passage from the supported nonzero word to full chain connectedness, and then to bounded degrees at *every* minimal horizontal step. The strict degree cutoff and the final induction are supporting deductions. The equality case is S1 itself. Neither the full named classification nor a literature search alone establishes novelty. Section 7 compares two recent primary proofs at the level of their operations.

2 Low degrees and closed incidence relations

A family of rational curves means a full irreducible component of $\mathrm{RatCurves}^n(X)$, as in [2, Definition 2.1]. It is *unsplit* if it is proper. Curves in the family are unparametrized irreducible rational images. We use the associated proper Chow family for chains.

Lemma 2.1. *Let X be Fano of pseudoindex i, and let γ be a positive numerical genus-zero stable-map degree with $d = -K_X \cdot \gamma < 2i$. Every genus-zero stable map of degree γ has a single nonconstant component, birational onto its image. Every full rational-curve family of that degree is unsplit. There are finitely many such full families, and the two-evaluation correspondence of degree γ is supported on their closed two-point incidence relation.*

Proof. Two nonconstant components would each contribute at least i to the anticanonical degree. A component covering its image with degree at least two would likewise contribute at least $2i$. Both are impossible. Forget all markings and stabilize. A contracted subtree would have a leaf with fewer than three special points, so no contracted component remains in the unmarked stable map. Its domain is $\mathbb{P}^1$ and it is birational to its image.

Properness of the unmarked stable-map spaces therefore allows neither a reducible limit nor a multiple-cover limit of curves of this degree. It follows that every full component of $\mathrm{RatCurves}^n(X)$ in the degree is proper. This is the source's minimum-degree incidence argument with its sufficient strict bound $d < 2i$ retained; it is not a new compactness theorem.

The fixed anticanonical bound gives bounded numerical coordinates, as in the source's setup. Its finite actual-degree convention gives a finite union of proper finite-type stable-map spaces. Each has finitely many components; passing to unparametrized images and their normalizations still gives finitely many full rational-curve components. Retain *all* of them. Their proper universal curves have proper two-fold fiber products, whose images in $X \times X$ form a closed set R_γ. Both evaluation points of a marked stable map lie on its unique rational image, even when a marking lies on a contracted bubble. Its evaluation pushforward is thus supported on R_γ. Every pair in R_γ lies on one of the retained curves. □

The associated Chow families introduce no extra chain relation: every cycle in these degrees is an irreducible reduced rational curve by the same degree argument. We do not select one component of an incidence space to represent a possibly signed virtual class.

3 A large chain support forces connectedness

For finitely many proper Chow families, we use the following precise rational-chain quotient contract. There is a dense open $U \subset X$ and a proper surjective morphism $\pi : U \to B$ such that each point fiber is an *entire* chain-equivalence class in X. In particular, U is saturated. After shrinking B and taking its inverse image, the fibers are pure of dimension $n - \dim B$. This is Kollár's theorem [4, IV.4.16], in the formulation of [2, Theorem 3.2]. The latter also gives a uniform chain bound $2^{n-\dim B} - 1$ within each fiber. Proper fibers are closed compact subvarieties of X; the quotient itself need not extend to X.

Lemma 3.1. *Let X be a smooth connected projective n-fold with $n \geq 2$. Let $\mathcal{V}$ be finitely many unsplit full families whose numerical classes span $N_1(X)$. Suppose a closed set $Z \subset X \times X$ consists entirely of pairs connected by $\mathcal{V}$-chains and satisfies $\dim Z \geq 2n - 1$. Then X is rational-chain connected by these families.*

Proof. Take the equidimensional saturated quotient just described, and put $b = \dim B$ and $E = X \setminus U$. If a pair in Z has one endpoint in U, its other endpoint lies in the same entire equivalence class, hence in U and in the same fiber. Consequently we have the actual containment

$$Z \subset (U \times_B U) \cup (E \times E). \tag{2}$$

The two constructible pieces have dimensions $2n - b$ and at most $2n - 2$. For $b \geq 2$ their union cannot contain a closed set of dimension $2n - 1$. Thus $b \leq 1$. The exceptional piece is $E \times E$; a one-endpoint exceptional locus would not give this conclusion.

Suppose $b = 1$. Choose one curve C_a from each family. Any C_a meeting U is wholly contained in a single fiber, because every point of the curve belongs to the same chain class. Choose a closed point $t \in B$ different from the finitely many resulting fiber values. Curves outside U also avoid

$F = \pi^{-1}(t)$. This nonempty proper fiber is closed in X and pure of dimension $n-1$. Any prime component D of F is a nonzero effective Cartier divisor, since X is smooth. It is disjoint from each C_a, so $D \cdot C_a = 0$ for all a. Spanning implies that D is numerically trivial. But for an ample divisor H one has $D \cdot H^{n-1} > 0$, a contradiction. Only finitely many representatives were avoided; numerical constancy does the remaining work.

Thus $b = 0$. The dense open U, as the fiber of a proper morphism to a point, is proper. Its inclusion in the separated variety X has closed image, so $U = X$. This fiber is a single entire chain class. The quotient theorem's uniform bound, now $2^n - 1$, also gives the bounded-chain convention of rational-chain connectedness used by Fujita. □

4 The composed virtual correspondence

By S1 and the hypothesis of Theorem 1.1,

$$0 \le \delta := n - r(i-1) \le 1.$$

The case $i = 1$ was excluded above. Put $s = \sum_j (d_j - 1)$ for S2. Then

$$r(i-1) \le s \le n, \qquad \sum_j (d_j - i) \le \delta \le 1. \tag{3}$$

Every $d_j \le i + 1 < 2i$. Lemma 2.1 gives a finite collection $\mathcal{V}$ of all full components in these degrees. Their numerical classes span $N_1(X)$, because they include the r independent degrees of S2.

We specify the intersection-theory construction and its support. For a cycle correspondence A on $X \times X$, use the action

$$A_*\alpha = (p_2)_*(p_1^*\alpha \cap \mathrm{cl}(A)).$$

If a source uses the opposite evaluation order, transpose its correspondence to adopt this convention. Composing A followed by B is the proper pushforward of the refined intersection on the matching middle factors; on a smooth triple product it is represented by $(p_{13})_*(p_{12}^*A \cdot p_{23}^*B)$. Its action is B_*A_*. Order the S2 factors accordingly (reading the operator word from right to left). Reversing this order convention changes none of the degrees or chain support arguments.

The standard refined Gysin construction is supported on the fiber product of the original supports. Proper pushforward, refined intersection, and the cycle-class map are compatible with this composition [5, §§6.2, 16.1, 19.1–19.2]. More explicitly, the product of the r supported virtual classes has dimension $rn + s$. Imposing the $r - 1$ matching diagonals of X, each of codimension n, produces a supported Chow class of dimension

$$rn + s - (r-1)n = n + s. \tag{4}$$

Push it forward to the two remaining endpoint factors. The intermediate support is the ordered incidence fiber product of the closed relations R_{γ_j}. All maps to the endpoint product are proper. Its image $Z \subset X \times X$ is therefore closed and consists entirely of pairs joined by actual $\mathcal{V}$-chains, including points on its boundary.

The endpoint Chow class lies in the image of $A_{n+s}(Z)$ and its cohomological action is the nonzero word in S2. It is therefore nonzero. If $\dim Z < n + s$, the group of cycles in dimension $n + s$ on Z is zero, so this would be impossible. Hence

$$\dim Z \ge n + s \ge n + r(i-1) \ge 2n - 1.$$

No effectiveness or positivity of virtual classes has been used. Nor have we closed a relation that was known only on general points: the chain support was a proper closed image from the start. Lemma 3.1 now proves that $\mathcal{V}$ rational-chain connects all of X when $n \geq 2$. The case $n = 1$ is already $\mathbb{P}^1$.

5 Bounded horizontal competitors

Fujita's Construction 3.5 starts with a minimal dominating family and then chooses a minimal horizontal dominating family for the quotient by the families already chosen. Minimality here and below is with respect to *anticanonical degree*. The construction ends with numerically independent families giving rational-chain connectedness [2, Definitions 2.1–2.2, Construction 3.5].

Lemma 5.1. *Suppose finitely many unsplit full families V_a rational-chain connect a positive-dimensional Fano manifold X, and each has anticanonical degree at most M. The initial minimal dominating family and every minimal horizontal dominating family in Fujita's construction have degree at most M.*

Proof. Every point lies on a curve from some V_a: connect it by a nonconstant chain to a different point. Properness makes each $\mathrm{Locus}(V_a)$ closed. Their finite union is X, so irreducibility makes one of them equal to X. This supplies a competitor for the initial minimum.

Consider an intermediate proper saturated quotient $\psi : U \to B$ with $\dim B > 0$. For a full irreducible family V_a, the condition that two points on a member lie in U and have distinct ψ-images is open on its two-fold universal curve. If nonempty it gives a dense constructible image, hence a nonempty open subset, in the irreducible parameter space: the general member is horizontal. If empty, ψ is constant on $C \cap U$ for every member C meeting U. The proper fiber is closed in X and contains this dense open subset of C, hence contains all of C. Thus a generically vertical family has no exceptional horizontal member.

For completeness, a vertical member meeting U cannot coexist with general horizontal members of the same proper family. That member lies wholly in a compact fiber and hence wholly in U. Properness of the universal curve makes the condition of lying wholly in U open around it. On this open family, the pullback of an ample divisor on the quasi-projective base has degree zero on that member and constant degree in the family. The curves therefore remain vertical. The two nonempty open conditions would contradict irreducibility. This also agrees with the usual horizontal-family convention in the cited construction.

Suppose none of the horizontal families dominates B. For each such family the closure of $\psi(\mathrm{Locus}(V_a) \cap U)$ is a proper closed subset of B. Choose $t \in B$ outside their finite union. No horizontal family meets $F = \psi^{-1}(t)$. Every member of the other families that meets F lies wholly in F, by the preceding argument. Induction on the length of a chain shows that every $\mathcal{V}$-chain starting in F stays in F. But F is nonempty and proper in X, contradicting the given full chain connectedness. Some V_a is horizontal and dominant over B, providing a competitor of degree at most M. Minimality gives the asserted bound. □

Apply the lemma to the collection from Section 4, with $M = i + 1$. Each family selected by Fujita has degree at most $i + 1 < 2i$ and is therefore unsplit by Lemma 2.1. This establishes the premise about *all* selected families in [2, Corollary 6.4]. We have not asserted that the original correspondence families all dominate X, or that they coincide with the newly selected families.

6 Classification and exact invariants

We prove Theorem 1.1 by induction on actual Picard number. The case $r = 1$ has $i \geq n$. The rank-one consequence of [3, Theorem 1.3] gives $\mathbb{P}^n$ or Q^n; Q^2 is excluded by its Picard number, and dimension one is $\mathbb{P}^1$. These are types (1) and (2). The equality case for every r is already S1.

Let $r \geq 2$ and suppose the statement holds for all smaller actual Picard numbers. Section 5 and Fujita's Corollary 6.4 give an extremal ray whose contraction has every fiber of dimension at most $i - 1$. The ray length is at least i. Wiśniewski's inequality and the maximal-length contraction theorem, as combined in [2, Theorems 5.1–5.2], show that the contraction is a $\mathbb{P}^{i-1}$-fibration. Its base Y is smooth and Fano, with

$$\dim Y = n - i + 1, \qquad \rho(Y) = r - 1, \qquad j := \iota(Y) \geq i \tag{5}$$

by [2, Lemma 4.2]. S1 ensures $n - i + 1 \geq (r - 1)(i - 1) \geq 1$, and

$$(r - 1)(j - 1) \geq (r - 1)(i - 1) \geq (n - i + 1) - 1.$$

The induction hypothesis applies at the actual invariants of Y.

These are precisely the base classifications used in the proof of [2, Proposition 5.3(b)]. That proof observes that the listed bases are rational, makes the projective-space fibration a projective bundle, and applies its Corollaries 4.6–4.7. It yields the five displayed types for X. This entire bundle classification is a cited result, not a new part of the argument.

There is no assumption that pseudoindex remains constant. In defect one, S1 on Y gives

$$(r - 1)(j - i) \leq 1.$$

Thus $j = i$ for $r \geq 3$. For $r = 2$ the base may be $\mathbb{P}^i$, of pseudoindex $i + 1$, or Q^i, of pseudoindex i; Q^2 is unavailable as a rank-one base. Fujita's Corollary 4.7(d) treats the $\mathbb{P}^i$ case and includes both nontrivial bundles.

One small point about the formulation of Proposition 5.3(b) deserves to be explicit. Its stated inductive premise is parametrized by a *lower* Picard bound. In its proof the needed base has actual Picard number $r - 1$, so the induction just given supplies the premise that is used. Alternatively, it supplies the whole parametrized premise: if a base has actual Picard number greater than the parameter, the arithmetic in the following paragraph and S1 identify it as a product of projective lines. No classification at a larger actual Picard number has been assumed in this induction.

To verify the equivalence with Fujita's parametrization, let $r_0 \leq r$ and $r_0(i - 1) \geq n - 1$. If $r > r_0$, S1 implies $(r - r_0)(i - 1) \leq 1$. Since $i \geq 2$, necessarily $i = 2$ and $r = r_0 + 1$. The two bounds then give $n = r$, so S1 gives $X = (\mathbb{P}^1)^r$. In the parametrized list this is $Q^2 \times (\mathbb{P}^1)^{r_0 - 1}$. Thus the actual-rank formulation loses no case of the advanced generalized Mukai classification.

We finish by checking the converse and all invariants. Projective space $\mathbb{P}^m$ has pseudoindex $m + 1$; Q^i for $i \geq 3$ has Picard number one and pseudoindex i. On products, Picard numbers add and the pseudoindex is the minimum of the factor pseudoindices: degrees of nonconstant projections are positive and add, and a line in a factor attains the minimum.

For $E = \mathcal{O}^{\oplus(i-1)} \oplus \mathcal{O}(1)$ on $\mathbb{P}^i$, put $\xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1))$ and let H be the pullback hyperplane class. The projective-bundle canonical formula gives

$$-K_{\mathbb{P}(E)} = i\xi + iH.$$

Both ξ and H are nef, and $\xi + H$ is ample. Each rational curve has positive integral $(\xi + H)$-degree, and a fiber line has degree one. This bundle is Fano of pseudoindex i, Picard number two, and

dimension $2i-1$. For $E = T_{\mathbb{P}^i}$ the same formula gives $-K_{\mathbb{P}(E)} = i\xi$. The tangent bundle of projective space is ample; ξ is ample and has degree one on fiber lines. The invariants are the same. Multiplication by $(\mathbb{P}^{i-1})^{r-2}$ gives types (3) and (4). Type (5) has pseudoindex i exactly because $r \geq 2$ supplies a $\mathbb{P}^{i-1}$ factor. All four defect-one types have dimension $r(i-1)+1$. Type (1) has dimension $r(i-1)$. This proves the converse and completes the proof of Theorem 1.1.

7 Comparison with related proofs

The pinned source's equality argument uses minimum-degree proper families, an ordered chain, and a chain-locus estimate. At equality the endpoint locus fills X; cyclic rotation makes the families covering and a product characterization finishes. With one unit of defect, that endpoint argument alone can give only a divisor-sized locus. It does not establish full chain connectedness or bound all the horizontal families in Fujita's construction. Here the dimension of the *composed cycle* is retained on the two-endpoint product. The saturated inclusion (2), the compact divisor contradiction, and Lemma 5.1 supply those missing conclusions.

Fujita's Theorem 6.1 assumes $r-1$ numerically independent dominating unsplit families. Its Corollary 6.4 instead assumes that every family in Construction 3.5 is unsplit. His Propositions 7.1–7.3 establish this in specified numerical ranges. We use the corollary at that exact hypothesis; neither the corollary nor its final induction is counted as an addition.

Watanabe's *Fano varieties of middle pseudoindex*, version 2 [6], proves the classification for $2i = n+1$ and $\rho(X) > 1$. This includes the rank-two defect-one case of our statement. Its proof treats birational and fiber-type contractions, uses one-point unsplit loci and intersections of fibers of distinct contractions, obtains a projective bundle, and applies existing bundle classifications. These are already established cases; we make no claim to newly classify them.

The revised version 2 of *Smooth Fano varieties with pseudoindex equal to half of their dimension* [7] assumes $i = n/2 \geq 2$ *and the existence of a birational extremal contraction.* It gives eight types under that hypothesis. Its proof separates small and divisorial contractions, uses unsplit loci and fiber-dimension inequalities to force another contraction to be a projective bundle, and invokes explicit bundle or blowup classifications. Its locus estimates include a needed irreducibility qualification for the initial subvariety. None of these arguments is an unrestricted replacement for the two-endpoint support and saturated-quotient argument here. The birational-contraction assumption must not be dropped when comparing that theorem.

These comparisons are based on reading the complete mathematical bodies of both versioned papers, including their proofs and references. They identify overlap and the different operation used here; they do not certify worldwide priority or constitute external peer review.

8 Source map and verification limits

The source is the manuscript directory `The-Generalized-Mukai-Conjecture-September-24-2026` in [1], at revision

`fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb.`

The exact source locations are:

- S1: `introduction.tex`, label `thm:main`;
- S2: `trace.tex`, label `prop:independent-correspondences`;
- S2's lower input: `spectral.tex`, label `prop:spectral`;

- correspondence conventions: `setup.tex`;
- degree-incidence comparison: `equality.tex`, label `lem:minimum-degree-incidence`.

The complete preprint tree, rather than a truncated repository tree, fixed the nine-file text, bibliography, and README inventory. The separately catalogued source `article.pdf` was not retrieved for that packet.

The following primary-reading record belongs to the original candidate author. The source's introduction, setup, trace, equality section, main file and relevant bibliography were read. Its spectral and descendant proof bodies were retrieved but not audited or substantively read: their outputs remain stipulated. Fujita's introduction and Sections 2–7 were read, including the full induction and bundle arguments. Dedieu–Höring's precise Theorem 1.3, discussion of the older gap, and proof overview were read; its full proof was not rechecked. The original proof of Kollár's quotient theorem was not newly read; its exact statement is used through Fujita. Fulton's cited sections identify standard refined-intersection, correspondence, and cycle-map contracts. A retrieved preview contained only front matter and contents, so those chapter bodies were not directly read in this preparation; the explicit construction in Section 4 states the compatibility being used. No stronger primary-reading claim is intended.

The mathematical draft and primary comparison were prepared by the author of the underlying proof packet. A successor implementation worker finalized formal-coverage wording, provenance, and artifact checks; that worker does not claim the original author's primary reads as its own. Separate internal proof and contribution reviews preceded this exposition; they do not constitute external publication review. Mathematical correctness is distinct from file integrity, compilation, and visual inspection.

Scoped formal support. A separate Lean package proves 29 supporting theorems among 35 public declarations. It covers finite degree budgets and strict numerical cutoffs; integer composition-dimension and quotient-dimension inequalities; the two-endpoint set inclusion from saturation and constancy for a map defined on the open subtype; preservation under actual reflexive-transitive chain closure; linear-functional vanishing on a spanning set and contradiction with a positive witness; and Picard, pseudoindex, and parameter arithmetic. The relevant saturation, support containment, numerical dimension bounds, spanning, positive witness, and generator-preservation properties are *supplied hypotheses* at these interfaces. Their geometric realization in this paper is not formalized.

The full five-type classification remains a human proof conditional on S1, essential S2, and the cited classical inputs. Lean does not construct Fano manifolds, rational-curve moduli, virtual cycles, refined intersections, proper equidimensional quotients, horizontal dominating families, Mori contractions, or the classified bundles. It does not verify their geometric dimension or positivity arguments or the converse invariant calculations. Neither S1 nor S2 is introduced as a Lean axiom.

The implementation uses Lean 4.34.1 and mathlib at revision

`d13f23b723b8a846827a245b89c10fc7d3f11612.`

A separate internal nonauthor audit returned `PASS_SCOPED_SUPPORT` with no required repairs after a fresh build and examination of every declaration and its exact type. Transitive axioms are confined to `propext`, `Classical.choice`, and `Quot.sound`. The implementation includes 11 proved control theorems, seven false-goal rejections, and four subprocess controls; the reviewer added nine boundary theorems and five false-goal tests. These checks concern the stated support only. They trust the installed Lean runtime and kernel and the pinned prebuilt dependency objects; the audit

did not bootstrap Lean or rebuild mathlib. The runner's source and text inventory was checked directly for this package and is not a general structured Lean-environment audit.

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References

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